

Advantages of using your own JavaScript Shell with GraalVM JDK Maven libraries.

This approach — rather than relying on the pre-built `js` launcher or Node.js runtime — allows you to create a powerful, customized, performance optimized, an highly integrated environment for executing JavaScript. At the same time, allowing seamless interoperability with Java.

- **Direct Access to Java APIs:** Your custom `jss`(JavaScript Shell) can instantiate Java classes, call Java methods, and access Java libraries directly from JavaScript without serialization or external processes. This is a core feature of GraalVM's polyglot architecture.
- **Unified Environment:** You can embed JavaScript logic directly within a Java application, allowing for dynamic scripting in a statically typed environment. This is particularly useful for applications that need to mix the flexibility of JavaScript with the robustness of Java. JSS is just a Java program consuming Maven dependencies, it can share the same classpath, configuration, logging, metrics, and dependency graph as the rest of the system.
- **Security Manager:** You can apply Java's Security Manager to restrict what the JavaScript engine can do (e.g., file system access, network calls), providing a controlled execution environment that is often superior to the default Node.js sandbox for enterprise use.
- **Standardized Build & Dependency Management:** Using Maven allows for standardized dependency resolution, reproducible builds, and easy integration with existing CI/CD pipelines that already manage Java libraries. This ensures that your shell is built with the same rigor and reliability as the rest of your Java ecosystem.
- **Resource Efficiency:** Applications compiled with GraalVM often require less memory and CPU compared to standard JVM applications. This leads to faster startup times, which is critical for serverless or microservices architectures, making them suitable for cloud-native and containerized environments.
- **As We Know, ECMAScript Compliance:** GraalVM provides a fully compliant ECMAScript runtime that supports modern JavaScript standards (up to ES2024), ensuring compatibility with modern JavaScript codebases.
- **No GraalVM Installation Required:** With the GraalVM JDK Maven libraries, the JavaScript engine ships as ordinary jars from Maven Central. Your shell can be built, tested, and deployed with a standard JDK and a normal `pom.xml`, removing the operational burden of installing a special JVM on every host.